# SHIHAB HOSSAIN

**Aspiring SQA Engineer · Manual & API Testing · Playwright · Python/Django**

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Portfolio: shihab-portfolio-frontend.vercel.app

## SUMMARY

Computer Science and Engineering graduate (Port City International University, CGPA 3.88/4.00, October 2026) focused on software quality assurance. I design test cases, test APIs with Postman and Newman, automate user journeys with Playwright, and write bug reports developers can reproduce. Testing my own Django application, I found and fixed 8 defects, including 3 security issues, and backed the fixes with 71 automated tests. Seeking an SQA internship or junior QA engineer role.

## SKILLS

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|---|---|
| **Testing** | Test case design, manual functional testing, bug reporting, regression testing, access-control testing |
| **API testing** | Postman (collections, environments, chained variables, assertions), Newman CLI with HTML reports |
| **UI automation** | Playwright Test (JavaScript), Page Object Model, Allure and Playwright HTML reports |
| **Database** | SQL, MySQL (relational schema design, constraints) |
| **Development** | Python, Django, Django test framework, C++, JavaScript, Git and GitHub, Docker (basic) |
| **Learning** | SDLC/STLC and test planning, CI test runs with GitHub Actions |

## QA PROJECTS

**DiagnoCare 2.0 — QA case study of a Django booking application** 2026

Manual testing · Security testing · Django test framework · Python github.com/shihab-hossain1/DiagnoCare_2.0

- Wrote 57 test cases across 7 areas (authentication, sign-up, booking, privacy and access control, centre workflow, reports, UI), each with steps, expected result and priority (32 × P1, 18 × P2, 7 × P3).
- Found and fixed 8 defects, 3 of them security issues: patients could see other patients' reports, any logged-in user could open any booking by changing the URL ID, and report PDFs were reachable by public URL.
- Covered the fixes with 71 automated tests (all passing); rewrote an assertion that passed for the wrong reason and confirmed the new tests fail without the fix.

**UI & API Test Automation — SQA course assignment (Ostad)** 2026

Playwright · Postman · Newman · Allure github.com/shihab-hossain1/SQA-shihab-assignment

- Automated 3 end-to-end scenarios on the SauceDemo store with Playwright: locked-out user error, 3-item checkout with subtotal, tax and total checks, and Z→A sorting with a computed total.
- Built a Postman collection of 9 requests with 19 assertions against the reqres.in API, including negative cases (400, 404), and ran it from the CLI with Newman. Recorded run: 3/3 UI tests and 19/19 assertions passed.
- Refactored the UI tests into a Page Object Model version (github.com/shihab-hossain1/sqa-automation-assessment).

**Patient Record Management System — relational database** 2025

MySQL · SQL · ER design github.com/shihab-hossain1/Database_Project

- Designed a 12-table schema for hospitals, doctors, patients, diagnoses and operations; primary keys, 13 foreign-key constraints and NOT NULL rules reject invalid or orphaned records.

**Other projects:** Bangladesh House Price Predictor (Python, machine learning, 2026) · Solar Tracking System (Arduino Uno, LDR sensors, C++, 2024)

## RESEARCH

- **B.Sc. thesis (completed, 2026):** Deep Learning-Based Traffic Accident Detection and Collision Classification from CCTV Videos — with Md. Ibrahim Khalil, supervised by Nusratul Jannat, PCIU. Two-stream ResNet50 + BiGRU/BiLSTM; 50.0% accuracy on the official real-video test set, above the best published baseline (47.1%); locked test set and fixed seeds for reproducible evaluation.
- **Conference paper, co-author (under review, 2026):** explainable machine learning for mental-health screening with minimal features — built the React and FastAPI web application.
- **Conference paper (under review, 2026):** system design of DiagnoCare, a web-based diagnostic test booking and report management system.

## EDUCATION

**B.Sc. in Computer Science and Engineering — Port City International University** 2022 – Oct 2026

CGPA 3.88/4.00. Coursework: Software Engineering, DBMS, Computer Networks, Web Technology.

HSC — Hazera Taju Degree College, Chattogram · GPA 5.00/5.00 2021

SSC — Nasirabad Govt. High School, Chattogram · GPA 5.00/5.00 2019

## TRAINING & ACTIVITIES

- Software Quality Assurance course — Ostad (in progress, 2026) · Programming course — Phitron (completed 2025)
- Member, PCIU Computer Club (2023 – present) · CSE Club coding contest and workshop (2026) · Peer tutor for school and college students (2019 – present)
- Languages: Bangla (native), English (Saifurs English course; Speakers Council spoken English certificate, 2019), Hindi